EC466 - Software Defined Networks

SELF-X

Ganesh: Submission Report

T20-T25 | IIIT Dharwad | Team SDN_07 | 06 November 2026

Submission status	Verified scope
READY - GANESH COMPONENT	CPU-trained LoRA action policy, frozen-replay evaluation, audited activation and rollback.

What was delivered

The component turns approved completed network episodes into training data, fine-tunes real LoRA weights in a pinned pretrained language model, evaluates candidate policies against an incumbent, and activates only candidates that pass a frozen non-regression campaign. The approved bounded-adoption policy is active in its local registry.

Task	Completed implementation / retained evidence
T20	Replay curation; source/split hashes; independent seed groups; original-input reconstruction; excluded-case audit.
T21	Pinned CPU model backend, constrained action tokens, local structured interface and independent action safety.
T22	One actual 120-step LoRA fit; 460,800 trainable parameters; frozen base verified after reloading saved weights.
T23	Frozen 576-case stress gate, raw metric replay, training lineage checks and fresh LM decision verification.
T24	Promotion, rejection, unchanged pointer on rejection, controlled failure rollback and hash-linked registry audit.
T25	Sealed six-seed holdout attribution with identical prompts and memory disabled; 24 on-policy episodes for a later cycle.

Research boundary

All network outcomes are calibrated fluid-model or frozen-replay measurements. The selected policy uses a 25% observation-hash rollout threshold for adapted weights and the pinned bare base elsewhere. Live Mininet integration, full governance/DPP coupling, A1-A5 evaluation and topology transfer remain separate team tasks.

Data and model record

Training/stress/holdout: 1,122 / 576 / 864 episodes. Thirty rejected training cases are retained. Base: HuggingFaceTB/SmolLM2-135M-Instruct, revision 12fd25f77366fa6b3b4b768ec3050bf629380bac. Only LoRA q_proj/v_proj matrices were optimized.

Verification and corrections

Check	Result
Project regression tests	106 main tests passed plus 4 subtests: 77 selfx/evaluation, 14 slow-loop and 15 Web.
Clean environment	22 focused tests passed in a new Python 3.12 environment; pinned dependencies and pip check passed.
Relocated package	22 focused tests, model reload verification and a 12-window backend smoke run passed with original project reads guarded. No forbidden reads or cloud inference calls.
Replay provenance	All three original replay inputs reproduce the curated train/stress/holdout files and excluded-case audit exactly. Portable source map retained.
Actual model decisions	5,184 decisions checked with fresh inference; no retained cache used. All actions matched. Maximum probability difference 1.19e-07 (tolerance 2e-05).
Tensor integrity	Reloaded frozen-base parameter hash matches the pre-training hash. Reloaded adapter hash matches the trained artifact and differs from initialization.
Lifecycle evidence	Accepted bounded policy; rejected weak checkpoint; rejected full-adapter trial; controlled automatic rollback; approved policy restored.

Issues corrected before submission

Malformed active-pointer files now remain inside the explicit no-op/rollback boundary. Rollback predecessors are checked against authorized activation history. Completed rewards and outcomes are independently recomputed. Promotion verifies action probabilities, exact installed decisions, provenance and dataset lineage. Source maps remove reliance on historical absolute replay locations.

Retained verification files

submission_verification.json contains the fresh inference and source-rebuild results. submission_readiness.json links the passed tests, portability checks and 44 verified source-file hashes. submission_validation/ contains machine-readable test results and relocated-run outputs.

Measurements and attribution

Holdout metric	Bare base	Bounded LoRA adoption
Cumulative reward, six seeds	64.188490	70.308932
Mean reward / observation	0.074292	0.081376
Modeled latency (ms)	19.014259	18.061938
Total loss (%)	4.083811	4.117937
Throughput (Mbps)	13.555983	13.553564
Unsafe proposals vetoed	484	353
Installed safety violations	0	0

Independent seed-level result

Across six independent seeds and three profiles, mean cumulative reward gain per seed was **1.020074**; 95% CI [0.675774, 1.364374]. Two-sided paired t-test: **p = 0.0006203**. Both conditions used the same frozen prompt and memory was disabled. The only learned difference was the LoRA weights within the fixed rollout mask.

Tradeoffs and selection record

Modeled latency and reward improved; loss rose by 0.034125 percentage points and throughput fell slightly. The first full-adapter candidate improved reward but exceeded the unchanged 0.10-point loss-regression allowance and was rejected. The accepted 25% hash-threshold variant selected 136/576 stress and 231/864 holdout observations for adapted weights. Holdout data did not select the deployment variant.

Interpretation

The result supports bounded weight-adaptation benefit relative to the pinned bare base in this replay model. Learning imitates approved bootstrap behavior. It does not establish superiority to the bootstrap heuristic, full-system A1-A5 significance, live packet performance or unseen-topology transfer.

Evaluation convention

Candidate and incumbent act on the same frozen observations and queue states. Outcomes are independently simulated for each decision; this is not a sequential deployment run. Independent seeds are the statistical replication unit.

Submission and reviewer quickstart

Artifact	Purpose
Ganesh_Submission.zip	Component source, adapters, replay inputs, evaluation traces, tests, pinned environment, reports and integrity manifest.
Ganesh_Offline_Model.zip	Optional pinned base-model companion, publisher model card, Apache license and model-file checksums. Extract beside the component archive.
SHA256SUMS.txt	Release checksums for both archives and this report.

Run from a new directory

Extract to a short path. From Ganesh_Submission/, use Python 3.12. Python packages require installation; the optional model companion avoids downloading the base model. No API credentials are needed.

```
python tools/verify_bundle.py
python setup_submission.py
cd selfx
.venv\Scripts\python.exe -m pytest tests/test_self_improvement.py -q
.venv\Scripts\python.exe -m scripts.verify_self_improvement --output tmp/check.json
```

On Linux/macOS, replace .venv\Scripts\python.exe with .venv/bin/python. The root README includes fresh-inference verification, backend smoke execution and new-cycle training commands.

Evidence directory

selfx/output/self_improvement/ganesh_20261106/bounded_adoption/ contains candidate/, registry/, dataset/, promotion_stress/, rejection_stress/, heldout_attribution/, rollback_demo.json, completion.json, submission_verification.json and submission_readiness.json. Original replay files, source_map.json and the rejected full-adapter evidence are retained in its parent folder.

Integrity and reproducibility

Archive paths and CRCs are validated, files have SHA256 manifests, and the packaged implementation must match the 44 source hashes that passed relocation. API keys, .env files, virtual environments, Git state and unrelated student artifacts are excluded. Historical absolute paths in immutable manifests are preserved as provenance; runtime loaders and source maps resolve packaged files.

Team handoff

The accompanying pending-work audit records Ganesh T20-T25 completed in CPU/frozen-replay scope and preserves the other 33 assignments. The overall course submission still requires those owners to complete their live integration, evaluation and final-system deliverables.